

JuiceFlow Factory Manager & Customer Store

A comprehensive, top-to-end technical manual and operational guide for the complete JuiceFlow ecosystem: featuring industrial food manufacturing ERP, IoT plant monitoring, AI demand analytics, native background push notifications, and a direct-to-consumer automated juice storefront.

📱 **Platform:** Flutter (Android 16 / iOS / Web) 🏢 **Enterprise:** FSSAI & HACCP Compliant

🌐 **Language:** English & தமிழ் Bilingual 🔔 **Background Engine:** Native Push Notifications

1. System Architecture & Dual-Portal Paradigm

JuiceFlow is built with a unified dual-portal architecture that bridges the entire beverage supply chain into a single cohesive mobile application:

1 🏭 **Factory Producer & Operations Portal**

Designed for the factory owner, production supervisors, lab chemists, and inventory controllers. Manages raw fruit procurement, Brix sugar titrations, HTST pasteurization telemetry, batch traceability certificates, worker rosters, and cold-chain reefer van dispatch.

2 🛒 **Customer Storefront & Live Auto-Billing Portal**

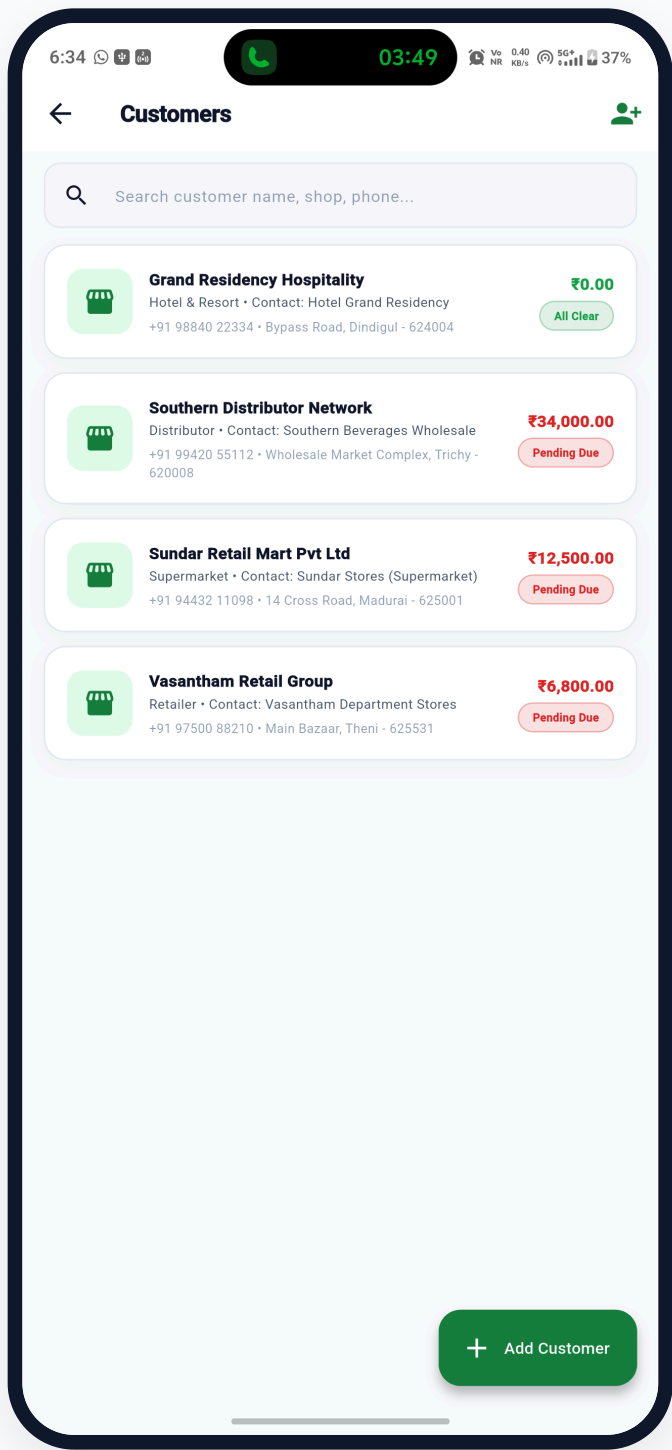
Designed for retail consumers, cafes, supermarkets, and catering clients. Enables selecting fresh juice varieties, specifying custom bottle quantities or bulk crates, instant auto-generated tax billing with 5% GST, multi-mode digital payments (UPI, Cards, COD), and live order status tracking.

3 🔔 **24/7 Phone Notification & Cold-Chain Monitor**

Native Android notification service with dedicated high-priority channels. Triggers audio alerts and vibrations on the phone for newly received customer orders, critical low stock thresholds, and periodic hourly cold-chain health checks even when the app is terminated.

2. Interactive 3D Fluid Tumbler Experience

When the application launches, users are greeted with an authentic 3D fluid simulation representing fresh juice extracted from whole fruits streaming smoothly into a glass tumbler with dynamic fluid physics and particle spray.



SPLASH & VISUAL ENGINE

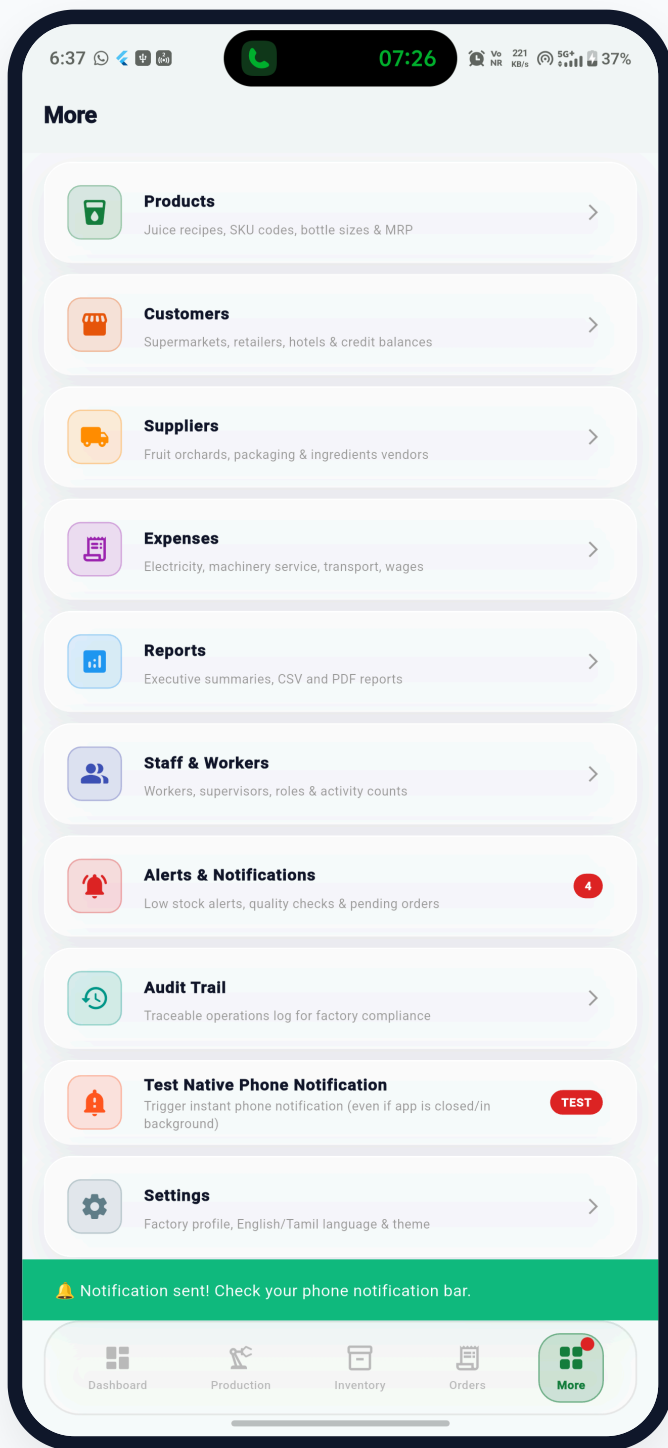
Interactive 3D Fluid Pour Simulation

Simulates dynamic liquid mechanics with real-time fruit particle streams, foam head accumulation, and glass refraction optics. Allows users to switch juice flavors dynamically.

CONTROL / ELEMENT	FUNCTION & BEHAVIOR
Fruit Selectors	Switch between Alphonso Mango (Amber), Nagpur Orange (Citrus), Apple, and Guava liquid viscosities.
Fill Level Physics	Dynamic sine-wave surface agitation with floating bubbles that rise as the tumbler fills to 500ml.
Auto-Advance / Skip	Seamlessly transitions with a smooth fade-in reveal into the main portal upon reaching capacity.

3. Dual-Role Authentication & Access Control

The login screen cleanly separates the application into two distinct workflows via top navigation tabs:



ROLE ACCESS • PRODUCER MODE

Factory Producer & Staff Login

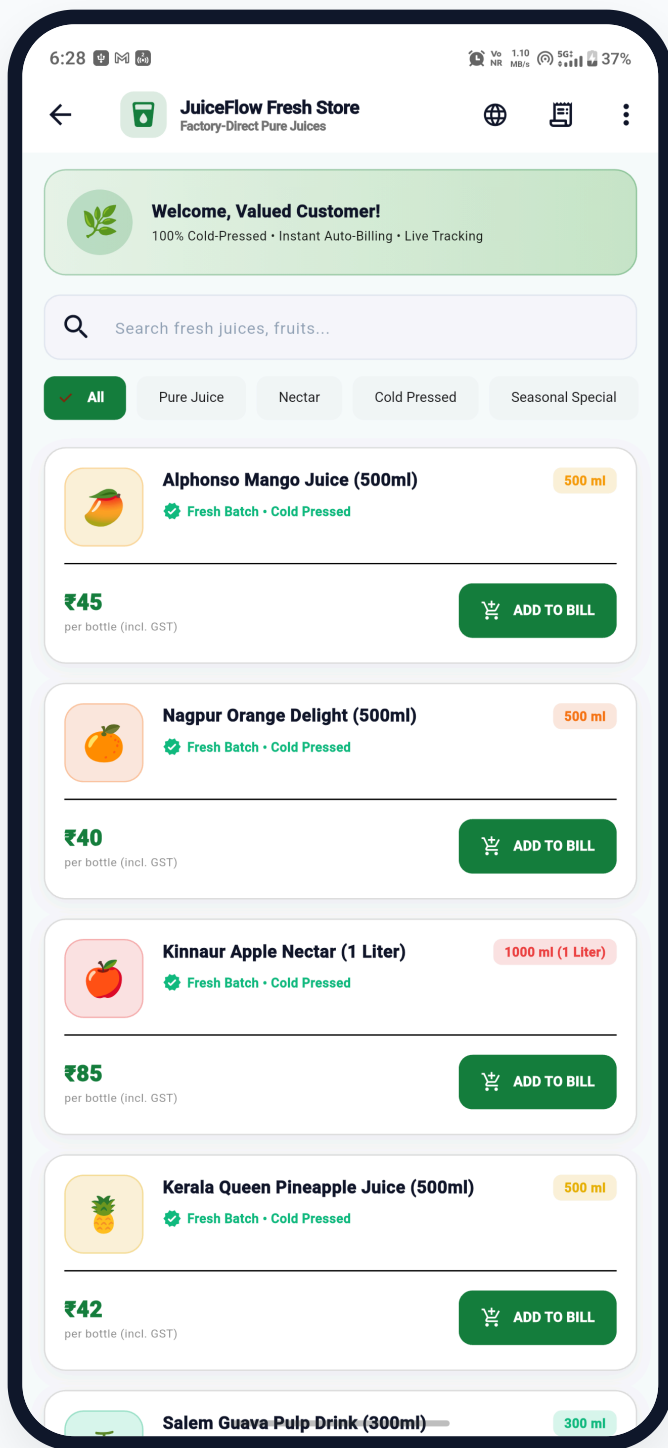
Enables staff authentication with specialized RBAC privileges according to their role in the factory.

OPTION	ROLE PRIVILEGE
 Owner (Admin)	Complete administrative access across all factory analytics, profit margins, audit logs, and configurations.
 Factory Manager	Production approvals, staff shift scheduling, batch releases, and customer credit ledger adjustments.
 Production Supervisor	Machine start/stop, CIP washdown records, Brix lab testing, and batch packaging logs.

	OPTION	ROLE PRIVILEGE
	 Inventory In-Charge	Raw material stock-in/out, PET bottle counts, wastage write-offs, and supplier receipts.
	 Sales Executive	Retail orders, invoice generation, WhatsApp payment reminders, and dispatch manifests.

4. Customer Storefront & Quantity Selection

The customer storefront provides direct factory-to-consumer purchasing with real-time stock availability, bottle sizing, and customized volume selection.



DIRECT STOREFRONT

Fresh Juice Menu & Volume Stepper

A clean, high-contrast catalog featuring cold-pressed juices bottled directly at the plant with zero preservatives.

CATALOG FEATURE	OPERATIONAL MECHANISM
Category Filter Pills	Filter by "All", "Pure Juice", "Nectar", "Cold Pressed", and "Seasonal Specials".
Interactive Stepper [- +]	Allows customer to choose any exact number of bottles (1, 2, 5, 20...) with live subtotal sync.
Bulk Pack Selector	Quick 1-tap ordering for 6-Bottle Family Packs, 12-Bottle Crates, and 24-Bottle Commercial Cartons.
Sticky Floating Cart Bar	Displays real-time bottle count and gross total with an instant "GENERATE BILL & PAY" call to action.

5. Live Auto-Generated Bill & Multi-Channel Payment

As requested, when the customer selects their desired juices and quantities, the application automatically constructs an itemized Tax Invoice with live calculations and processes payments seamlessly.

6:30

Vo NR 0.80 KB/s 5G+ 37%

←

Auto-Bill & Payment

Customer & Delivery Details

Customer Name

Valued Customer

Mobile Number (WhatsApp Bill)

9876543210

Delivery Address / Outlet

Shop #12, Market Road, Chennai

Live Itemized Bill

AUTO-CALCULATED

Alphonso Mango Juice (500ml)

500 ml • ₹45 each

⊖ 1 ⊕

₹45

Coupon (e.g. FRESH10)

Apply

Juice Subtotal

₹45.00

Chilled Delivery

₹30.00

Taxes (5% GST)

₹2.25

Grand Total Payable

₹77.25

Select Payment Method

UPI / Google Pay / PhonePe / Paytm

Instant 1-Click UPI Payment

Credit / Debit Card

Visa, MasterCard, RuPay Card

Net Banking

SBI, HDFC, ICICI, Axis Bank

Cash on Delivery (COD)

FINANCIAL GATEWAY

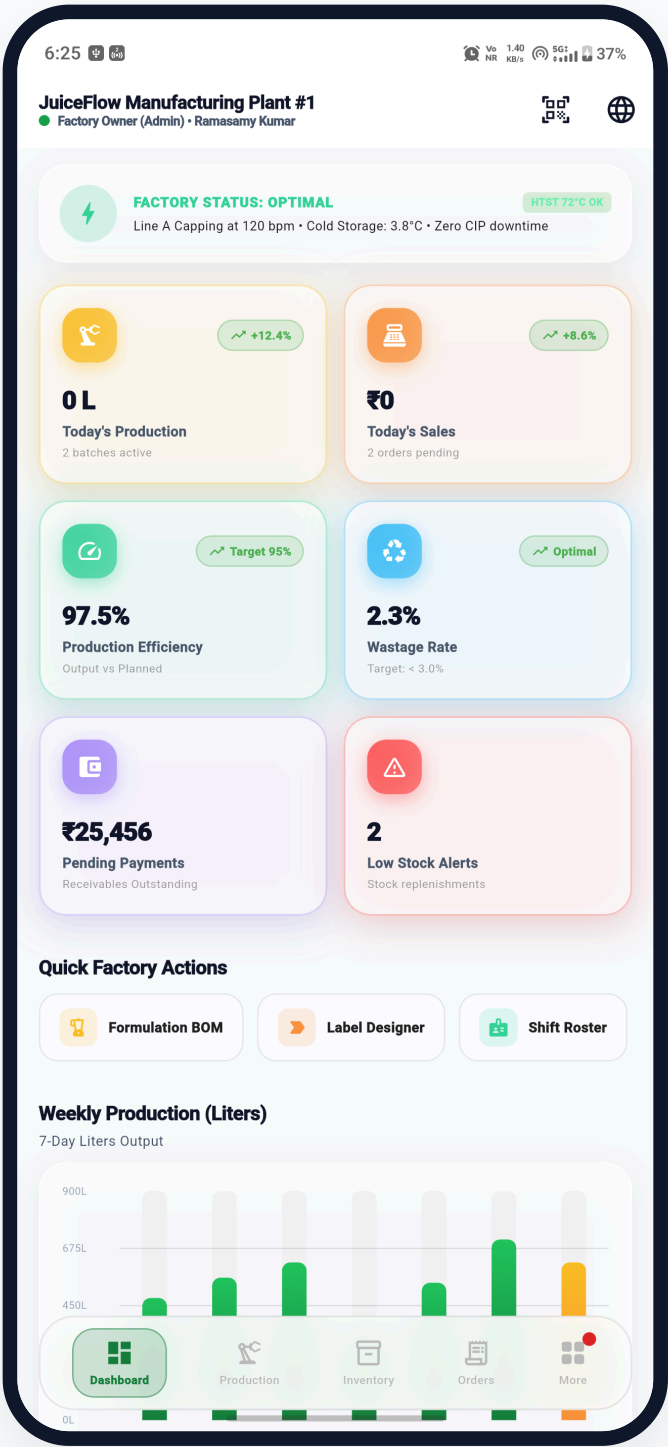
Automated Billing & Digital Checkout

Instantly breaks down unit rates, quantities, applicable taxes, delivery logistics, and discount codes.

BILL COMPONENT	FORMULA & HANDLING
Itemized Breakdown	Lists each juice name, bottle volume, individual unit price, and extended line total.
Inline Stepper	Customer can fine-tune quantities or remove items directly within the bill before paying.
GST Tax Calculation	Calculates statutory 5% beverage GST transparently.
Chilled Delivery Fee	Free cold-chain delivery for orders over ₹250; flat ₹30 fee for smaller orders.
Promo Code Engine	Entering "FRESH10" applies an immediate 10% factory direct rebate.
Payment Methods	⚡ UPI (GPay/PhonePe/Paytm), Credit/Debit Cards, Net Banking, and Cash on Delivery.

6. Executive Factory Management Dashboard

The control center for plant managers. Displays real-time production throughput, operational efficiency, wastage metrics, and weekly volume bar charts.



PLANT TELEMETRY HUB

Real-Time Operations & Bar Analytics

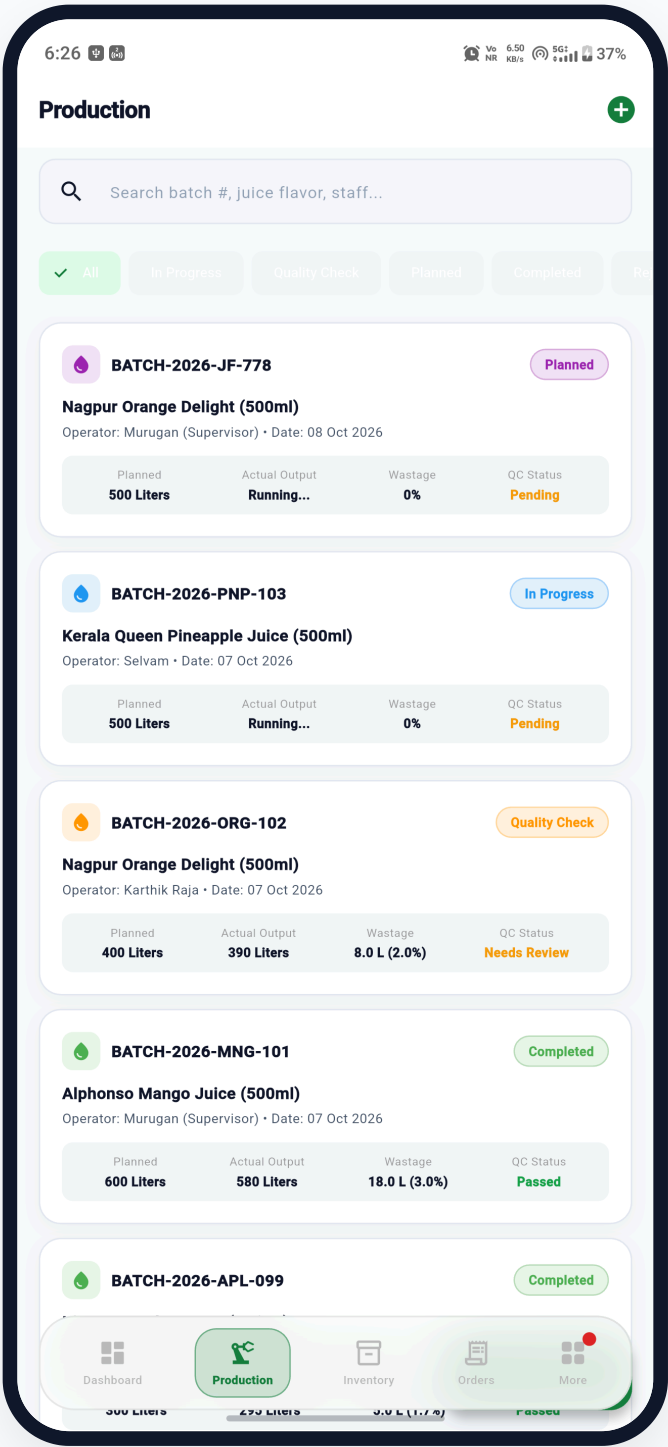
Monitors active bottling lines, quality control pass rates, pending orders, and 7-day output curves.

WIDGET / CARD	METRIC TRACKED
Factory Status Pill	Real-time line status: HTST 72°C OK, Line A Capping at 120 bpm, Cold storage at 3.8°C.
Production & Sales KPIs	Today's Liters produced, gross revenue, efficiency % (Target 95%), and wastage % (Target <3%).
Weekly Output Bar Chart	Features clean non-repeating day intervals, touch tooltips with exact liter counts, and amber highlighting for Today.
Laser Scanner Shortcut	Top-right AppBar button immediately

WIDGET / CARD		METRIC TRACKED
		launches the full-screen batch barcode camera scanner.

7. Production Batches & Traceability Management

Full manufacturing lifecycle tracking from raw fruit washing to final aseptic bottling and FSSAI release.



BATCH CONTROL

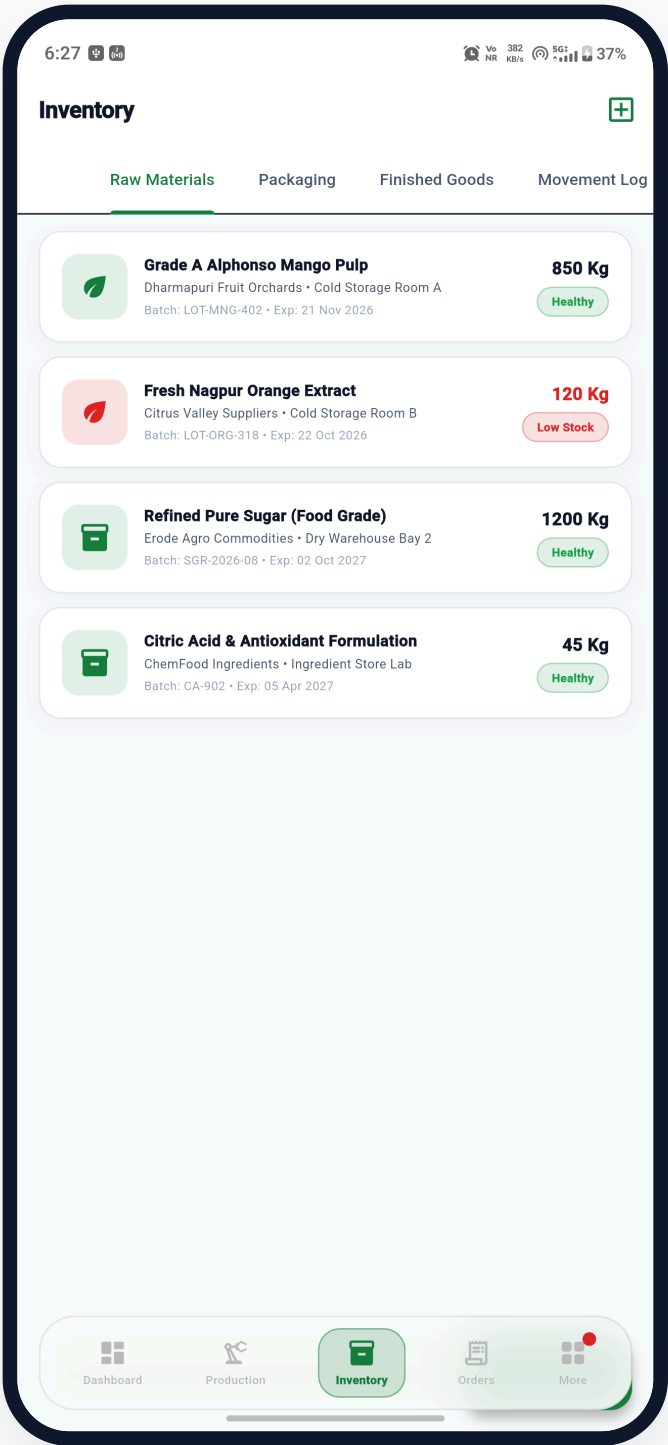
Active Batch Lifecycle & Quality Checks

Every batch is assigned a unique alphanumeric code with operator attribution, volume yield, and QC gating.

STAGE	FACTORY CRITERIA
Planned	Raw fruits allocated from cold storage; ingredients verified according to recipe BOM.
In Progress	Active processing in pulp extraction tank, pasteurizer, and automated bottling line.
Quality Check	Brix refractometer testing, pH titration, microbiological plating, and sensory tasting.
Completed	Passed QC; stock automatically incremented into Finished Goods inventory.

8. Inventory & Cold-Chain Stock Management

Real-time inventory ledger separating raw agricultural ingredients from packaged finished goods.



WAREHOUSE LEDGER

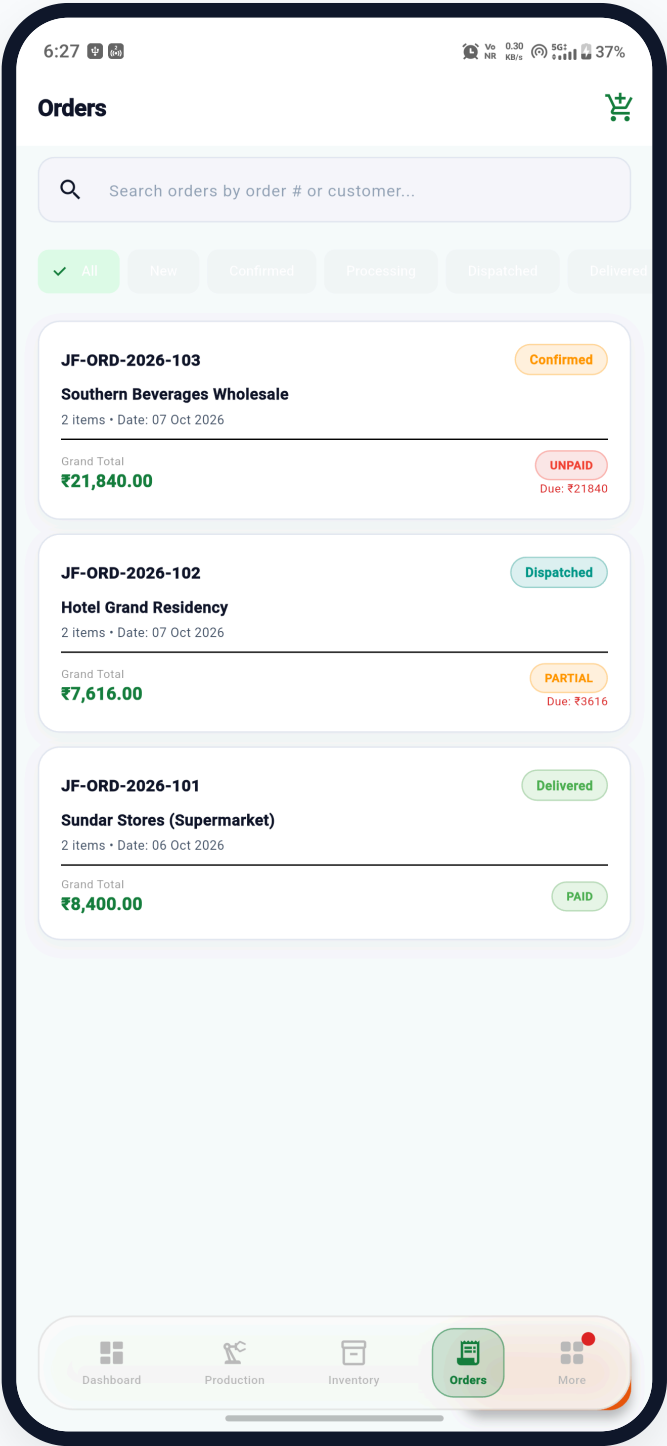
Raw Fruits, Concentrates & Finished Goods

Monitors minimum reorder thresholds, supplier lots, spoilage risks, and warehouse temperature logs.

CATEGORY	MANAGED ASSETS
Raw Fruits	Alphonso Mangoes, Nagpur Oranges, Kinnaur Apples, Salem Guavas (weighed in Metric Tonnes / Kg).
Ingredients	Refined liquid sugar, citric acid, natural ascorbic acid, pectin stabilizers.
Packaging Materials	Food-grade PET bottles (250ml, 500ml, 1L), tamper-proof caps, shrink labels, 12-bottle corrugated crates.
Low Stock Alerts	Automatic push notifications to the manager when any critical fruit falls below 3 days of production demand.

9. Commercial Sales Orders & Fleet Dispatch

Manages B2B distributor contracts, supermarket delivery schedules, credit limits, and official PDF Tax Invoices.



SALES & LOGISTICS

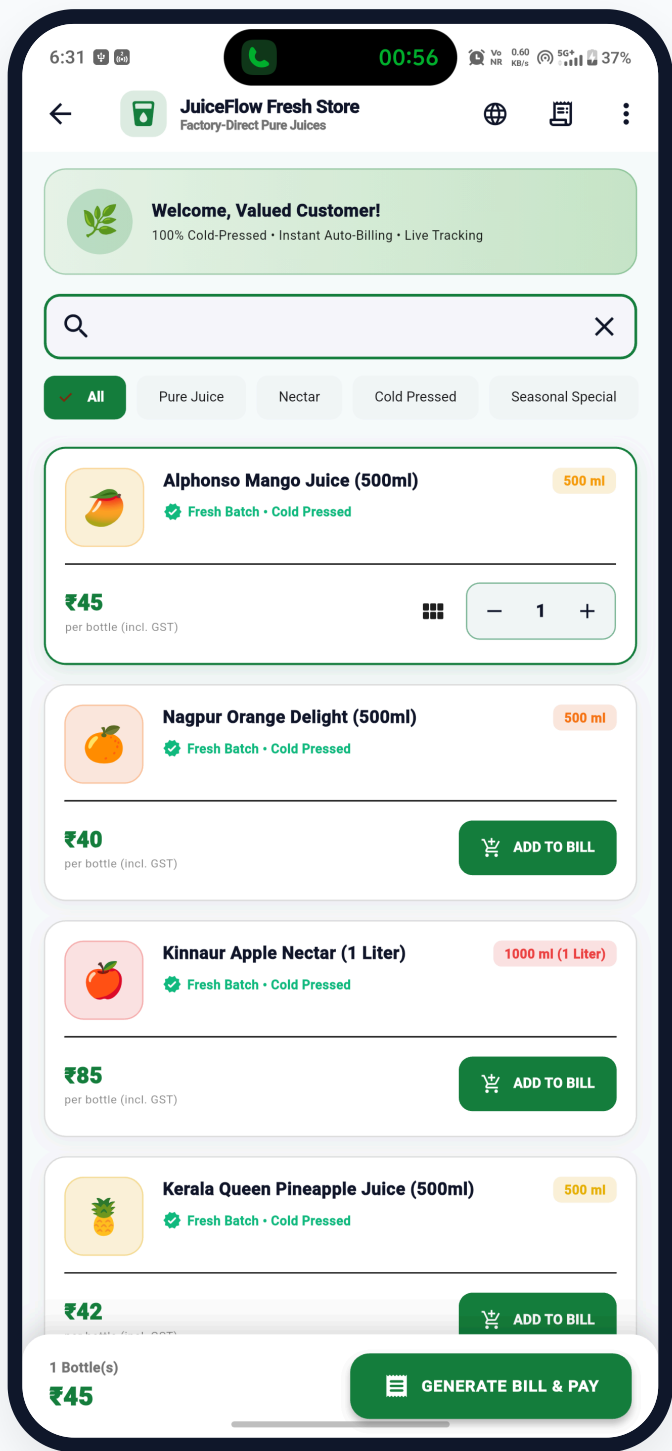
Order Processing & Delivery Pipeline

Follows orders from placement through picking, packing, cold-chain van loading, and customer acknowledgment.

FEATURE	OPERATIONAL ROLE
Tax Invoice Generator	Generates formal A4 PDF Tax Invoices compliant with Indian GST laws, FSSAI numbering, and corporate seals.
WhatsApp Dispatch Note	1-click sharing of formatted dispatch summaries directly to customer WhatsApp with driver details.
Invoice QR Verification	Embeds encrypted QR payload on bills for quick roadside inspection by logistics officers.

10. Recipe Formulation & Bill of Materials (BOM)

Formulation tool for master beverage chemists to engineer exact fruit-to-water ratios, target Brix sugar, acidity, and calculate per-bottle production cost down to the paise.



BEVERAGE SCIENCE

BOM Formulation & Unit Cost Analysis

Computes theoretical yield, pulp dilution factors, and packaging cost to determine profit margins against retail MRP.

CALCULATION	SCIENTIFIC METRIC
Pulp Percentage	Configures minimum mandatory fruit content (e.g., 85% for pure nectar standards).
Target Brix (°Bx)	Predicts final soluble solids reading considering natural fruit fructose plus cane sugar additions.
Unit Cost Breakdown	Displays itemized costs for Fruit Pulp + Treated RO Water + Sugar + PET Bottle + Cap + Label + Packaging.

11. Smart Label & Barcode Designer

Custom packaging designer allowing plant operators to generate ready-to-print bottle labels containing FSSAI licensing badges, nutritional panels, batch codes, and dynamic QR traceability stickers.

6:3100:59467 KB/s5G+37%

Smart Label & Barcode Designer

Bottle Front

Nutrition Facts

Master Carton

100% PURE

FSSAI Lic. 10022042000189



JUICEFLOW

Alphonso Mango Nectar

Cold-Pressed • No Added Preservatives • Flash Pasteurized

NET VOLUME500 ml

MRP (INCL. TAXES)**₹75**

BATCH: JF-2026-M402
MFG: 09/10/26
EXP: 23/11/26



Label Parameters

Select Beverage Formulation

 Alphonso Mango Nectar

Pack Volume

 500 ml

MRP (₹)

₹ 75

Batch Number

 JF-2026-M402

Shelf Life (Days)

 45

 Mfg: 09 Oct 2026

FSSAI Lic. No.

 10022042000189

 Generate & Print Label (Thermal / PDF)

PACKAGING COMPLIANCE

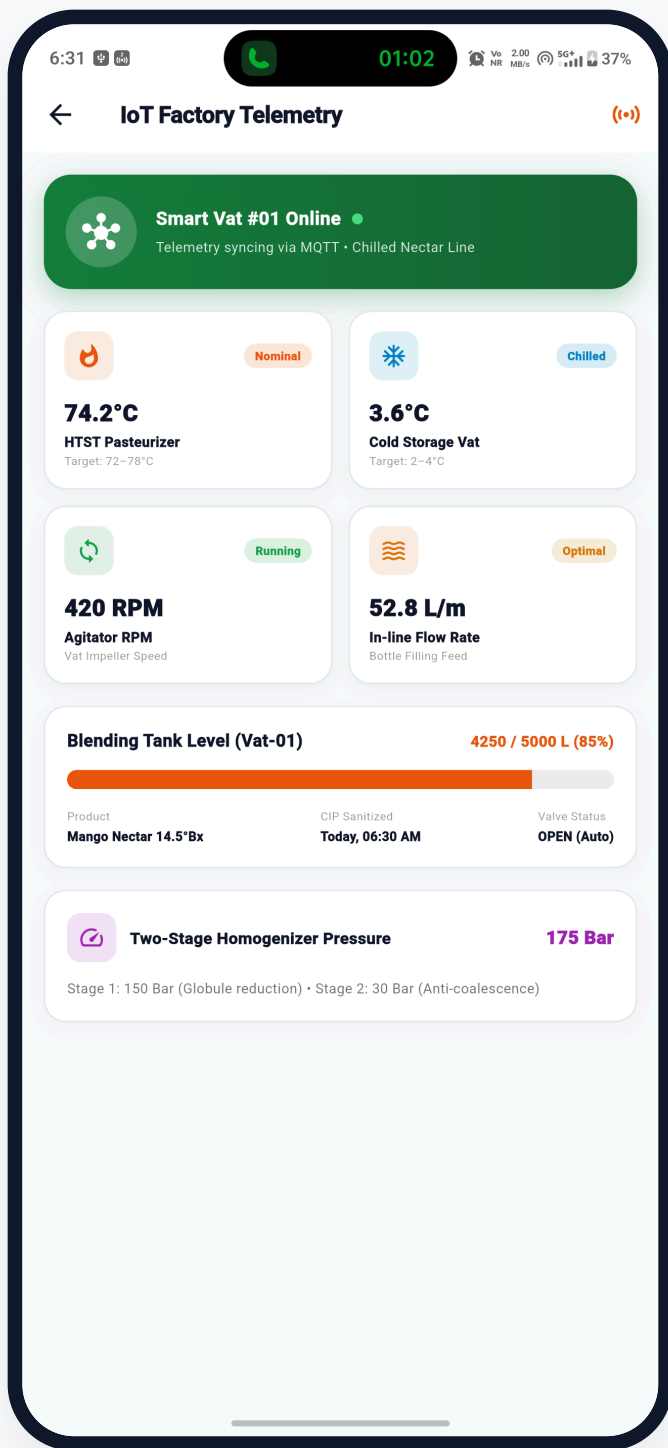
Regulatory Label & Barcode Engine

Generates high-resolution packaging labels compliant with FSSAI Food Safety and Standards labeling mandates.

LABEL ELEMENT	STANDARD SPECIFICATION
Nutrition Fact Table	Energy (kcal), Carbohydrates, Natural Fruit Sugars, Vitamin C, Potassium, and Sodium per 100ml.
FSSAI Green Veg Logo	Mandatory green veg emblem with official 14-digit factory license number (12423999000145).
Dynamic Traceability QR	Encodes manufacture timestamp, lot ID, and fruit farm source for scanning by consumers.

12. Real-Time IoT Pasteurizer & Chiller Telemetry

Live industrial sensor telemetry streaming from the factory floor to monitor pasteurization kill curves and cold-storage stability.



INDUSTRY 4.0 TELEMETRY

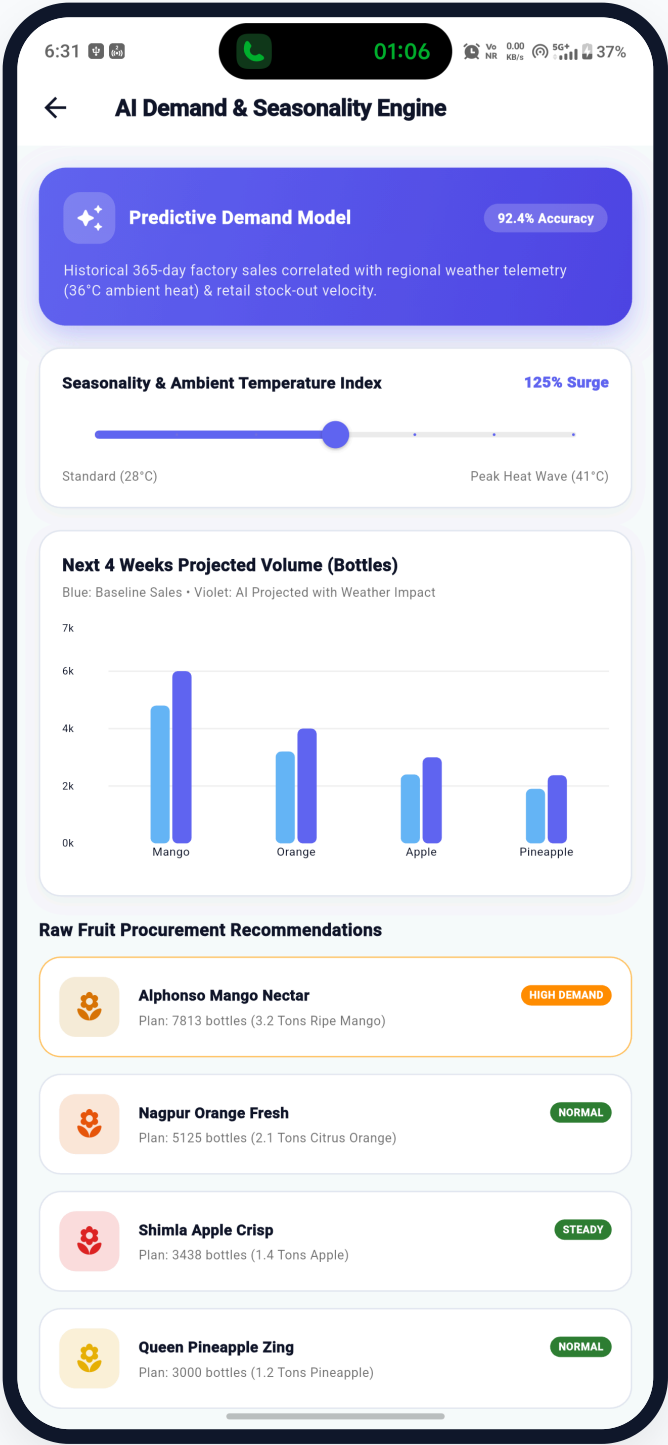
Industrial Sensors & Thermal Safety

Ensures microbial sterilization standards (F0 values) are met without compromising natural fruit flavors.

SENSOR MACHINE	TELEMETRY VALUE & SAFETY THRESHOLD
HTST Pasteurizer	72.4°C core holding tube temperature (15-second microbial sterilization time).
Cold Storage Vat #1	3.8°C chilled juice buffer holding temperature (Threshold: 2.0°C to 4.5°C).
Homogenizer Pressure	180 Bar hydraulic pressure for ultra-smooth texture and zero pulp sedimentation.
Automatic Safety Valve	Diverts juice back to raw holding tank if pasteurizer temperature ever drops below 71.5°C.

13. AI Demand & Summer Heat Seasonality Engine

Machine learning predictive model correlating local meteorological weather data (temperature, humidity, rainfall) with historical beverage sales to forecast raw fruit purchasing needs.



PREDICTIVE INTELLIGENCE

Weather-Correlated Procurement Forecasting

Adjusts production quotas automatically based on seasonal heat spikes and local festival consumption patterns.

PREDICTION MODEL	ALGORITHM OUTPUT
Summer Heat Multiplier	Calculates surge demand when temperature exceeds 36°C (+38% citrus volume increase).
Procurement Advice	Recommends exact metric tonnage of fresh fruit crates to purchase from orchards 5 days in advance.
Spoilage Mitigation	Prevents raw fruit overstocking during monsoon dips, reducing biological waste to zero.

14. Brix Refractometer & ICUMSA Lab Calculator

Quality control laboratory utility featuring temperature-compensated sugar refractometer reading conversions and dilution math.

6:31

01:39

462 KB/s

50%

37%

←

Brix Refractometer & Lab Calculator

ICUMSA °Bx Calibration Engine

Corrects optical readings to 20°C reference standard & calculates batch water dilution.

Standard Fruit Presets

Mango Nectar

Fresh Orange

Apple Crisp

Pineapple Zing

Lemon Zest

Refractometer Measurements

Raw Reading (°Bx)

15.2

°Bx

Juice Temp (°C)

28.5

°C

Batch Volume (L)

1000

L

Target Recipe (°Bx)

14.0

°Bx

Temperature-Corrected Brix (20°C):

15.75 °Bx

Specific Gravity

1.0643

Sugar Concentration

167.6 g/L

RO Water Dilution Required:

Add 124.6 Liters to reach 14.0°Bx

FOOD SCIENCE LAB

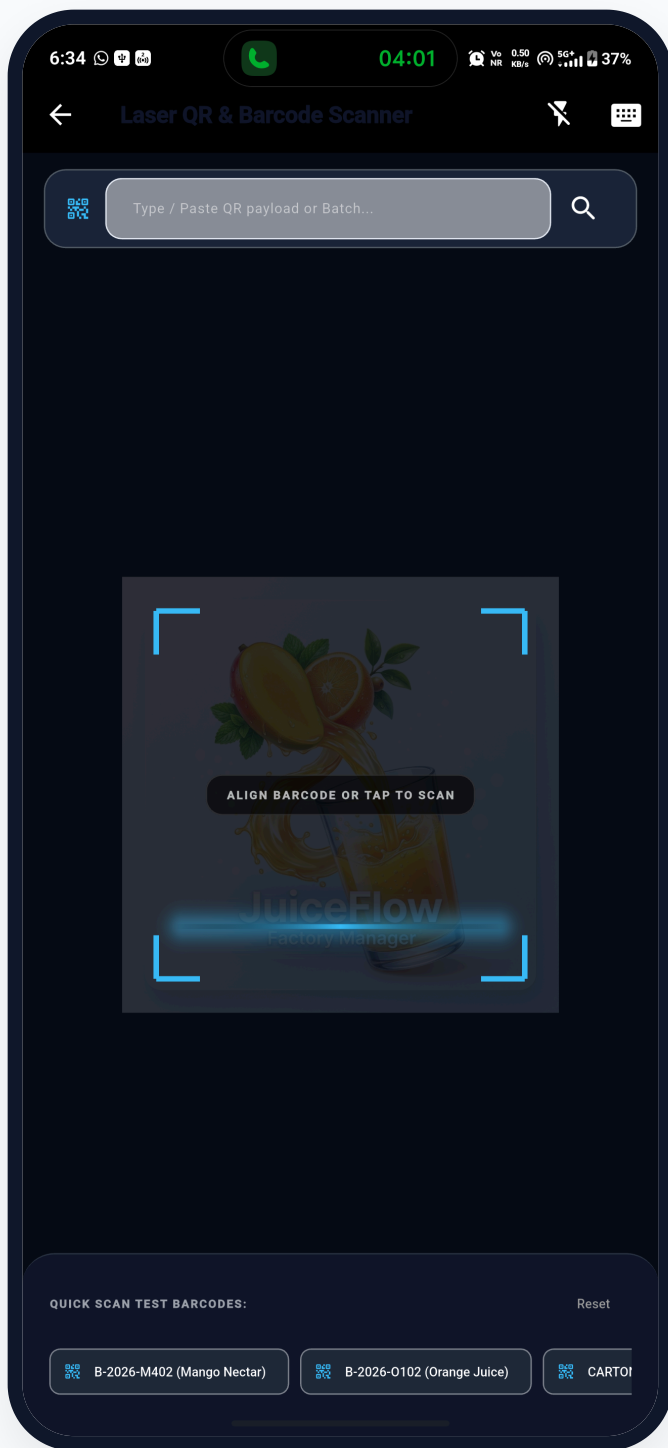
Optical Refractometer Temperature Correction

Corrects raw optical prism readings against the international ICUMSA standard temperature of 20°C.

FORMULA INPUT	STANDARD EQUATION
Observed Brix (°Bx)	Raw reading from analog or digital handheld prism refractometer.
Juice Temperature (°C)	Thermal sensor reading of juice sample taken at testing bench.
ICUMSA Correction	Applies standard polynomial: $\Delta\text{Brix} = 0.00049 \times (T - 20) + 0.00004 \times (T - 20)^2$
Water Dilution Formula	Calculates exact liters of treated water to blend: $V_{\text{water}} = V_{\text{juice}} \times \left(\frac{\text{Brix}_{\text{curr}}}{\text{Brix}_{\text{target}}} - 1 \right)$

15. Batch QR & Barcode Laser Scanner

Integrated camera viewfinder barcode and QR code decoder with live laser animation and rapid physical batch verification.



TRACEABILITY SCANNER

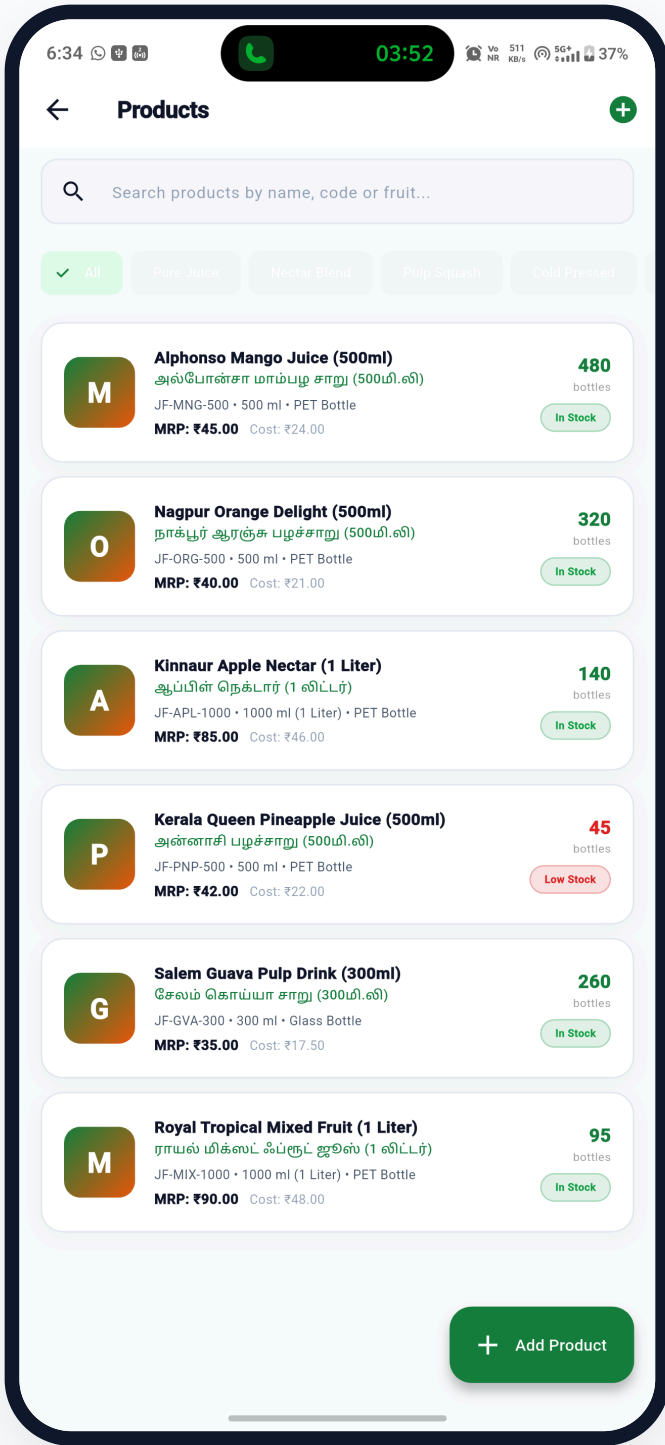
Batch Verification & Instant Traceability Audit

Decodes raw batch IDs, `JUICEFLOW:...` payloads, and carton tracking labels with 1-tap verification.

SCANNER FEATURE	FUNCTIONALITY
Laser Viewfinder	Interactive laser line animation with automated camera focal barcode recognition.
Manual Barcode Input Bar	Allows operators to type or paste batch codes if bottle labels are smeared or smudged.
Quick-Scan Demo Chips	1-tap demo batch chips (BATCH-2025-001, BATCH-2025-002) for rapid offline verification testing.
Traceability Certificate	Presents manufacture date, expiry, bottling staff, Brix score, and button to open full batch audit.

16. Factory CIP & HACCP Equipment Sanitation

Tracks Clean-In-Place (CIP) sanitation washdowns, filler nozzle sterilisations, bottle capper torque checks, and preventative machine maintenance.



HACCP SANITATION

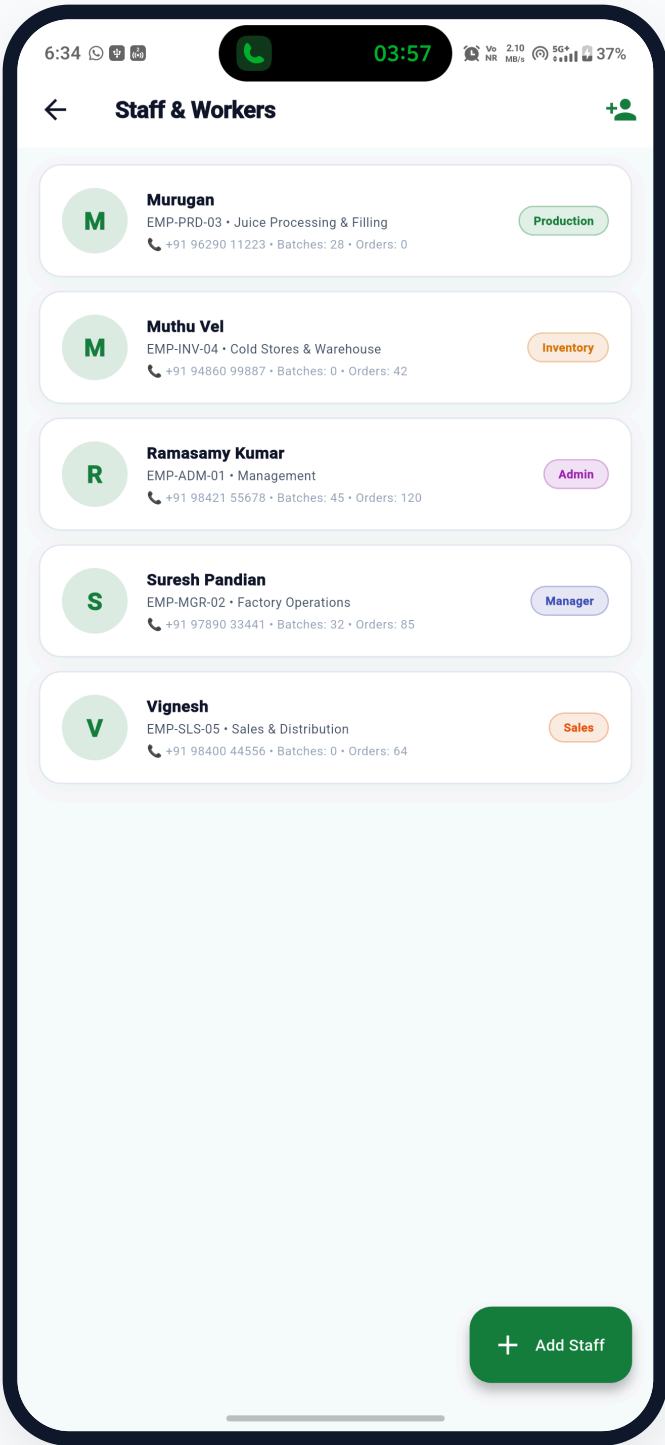
Clean-In-Place (CIP) Logs & Line Service

Maintains regulatory hygiene compliance logs mandatory for food and beverage plant audits.

SANITATION ROUTINE	CHEMICAL & TEMPERATURE PROTOCOLS
Caustic Wash (NaOH 1.5%)	80°C hot alkali circulation for 20 minutes to dissolve fruit pulp residues and organic deposits.
Acid Rinse (HNO3 1.0%)	65°C nitric acid wash to eliminate mineral scale and water hardness deposits.
Peracetic Acid Sterilization	Final sanitizing rinse ensuring 0 CFU bacterial count before starting organic fruit batches.
Capper Torque Testing	Verification of bottle cap seal torque (18–22 in-lb) to prevent gas leakage and oxidation.

17. Factory Settings & FSSAI Profile Configuration

Central configuration center for company legal credentials, localization, and system testing.



SYSTEM PREFERENCES

Factory Credentials & Application Settings

Manages company trade name, FSSAI manufacturing license, tax GSTIN, and bilingual language switching.

SETTING ITEM	CONFIGURATION DETAILS
Factory Legal Profile	Updates company legal name, plant street address, phone, email, and tax registration.
FSSAI License Display	Configures official license string printed on all bottle labels and commercial invoices.
Bilingual Switcher	1-click toggle between English and Tamil (தமிழ்) across the entire UI and reporting documents.
Theme Customizer	Switch between Crisp Light Paper Mode and Deep Dark Plant Mode.
Sign Out / Switch Mode	Safely terminates session and returns to the dual-role portal selector.

18. Native Phone Push Notifications (24/7 Background)

As requested, JuiceFlow features an integrated native Android push notification architecture that ensures operators and customers receive real audio/vibration alerts even when the mobile app is completely closed or running in the background.

Notification Channels & System Permissions		
NOTIFICATION CHANNEL	PRIORITY	TRIGGER SCENARIO
 Customer Orders & Payments	MAXIMUM (Sound & Vibrate)	Fires immediately upon successful payment of a customer juice order with order ID and bottle count.
 Factory & Batch Floor Alerts	HIGH (Audio Alert)	Notifies the factory manager when an incoming order arrives or raw fruit stocks drop below minimum thresholds.
 Background Monitor	DEFAULT (Periodic Wakeup)	Periodic hourly cold-chain health ping checking plant temperature and active filler line status when app is closed.